

Extending The MOBILE NETWORK FOR THE IoT



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With several billion devices expected to be connected to the Internet of Things (IoT) over the next few years, character of the mobile network will be changed completely. The types of devices involved broadly fall into two categories: massive machine-type communication (MTC) and mission-critical MTC. The demands these place on the network are very different from each other, as well as from those of traditional mobile terminals or user equipment. The resulting IoT ecosystem will be complex—multi-layered and cross-disciplinary as shown schematically in the figure.

Massive MTC

Protocols developed for mobile phones, smartphones and tablets are unnecessarily complex for high-volume non-speech devices that will be used for massive MTC, and these devices will also need to be low-power in order to have ultra-long battery life, and must have a low unit cost.

Massive MTC use cases include connected lifestyles and healthcare applications such as wearable technology, as well as smartenergy, public infrastructure and industrial automation. These applications generally have intermittent, low data rate requirements and no

particular latency constraints, so will not need the high data rates promised for 5G.

Nevertheless, the sheer volume of devices forecast to connect to the IoT means that eventually the capacity of 5G will be needed to support these all. In the meantime, new low-complexity protocols are being introduced for 4G, which will allow significant numbers of MTC devices to work on 4G LTE networks up to and beyond the launch of 5G.

3GPP LTE-M standard (also known as Cat-M or Cat-M1) will enable lower-cost, lower-power M2M devices to work within the currently available mobile networks, and will also offer modes that improve in-building coverage.

Two further IoT standards, enhanced coverage GSM (EC-GSM) and narrow-band IoT (NB-IoT), are also being introduced. EC-GSM, as its name suggests, uses legacy 2G technology, while NB-IoT can be used with either 4G LTE or for re-farming of GSM channels.

NB-IoT is a new narrow-band radio technology introduced by 3GPP to specifically address the requirements of the IoT. Standardisation began in September 2015 and is still ongoing. It is designed to improve indoor coverage as well as provide support for a massive number of low-cost, low-throughput devices with low device power consumption and optimised network architecture.

NB-IoT can either be deployed in-band, utilising resource blocks within a normal LTE carrier or its guard band, or in a standalone mode for deployments in non-LTE spectrum. It will entail a radical change to the physical layer, occupying a bandwidth of only 200kHz including guard-bands. This allows one GSM/GPRS chan-

Schematic of the proposed IoT ecosystem

